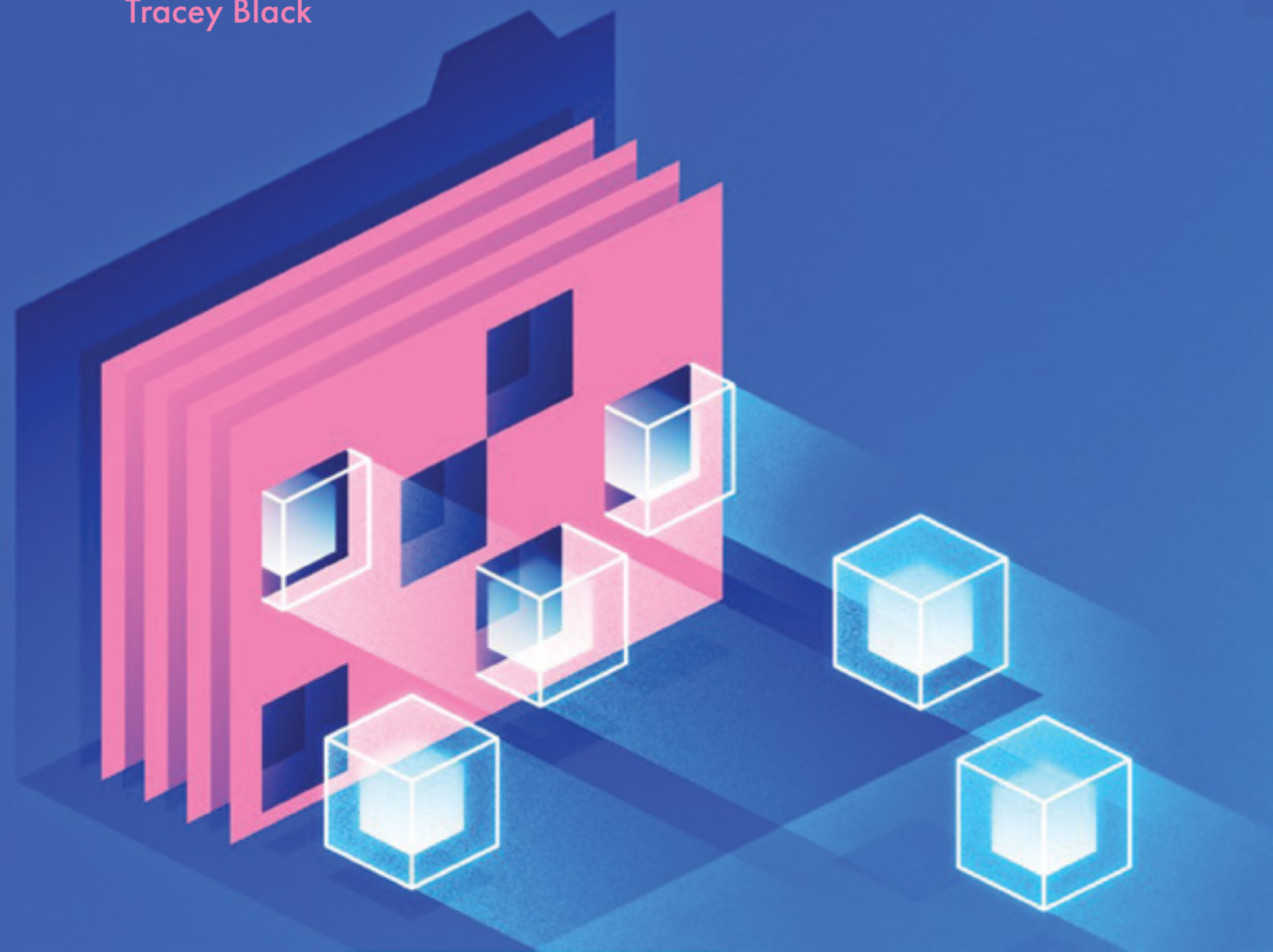
Centre for International
Governance Innovation

CIGI Papers No. 360 — August 2026

# Distributed Ledger Technology and the Future of Payments in Canada

Tracey Black



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## About the Author

Tracey Black is a recognized leader in payments, with experience spanning banking, consulting and national payments infrastructure. She served as president and CEO of Payments Canada, leading the operation and modernization of Canada's core payment systems. She previously held senior roles at Toronto-Dominion Bank, Royal Bank of Canada and McKinsey & Company. She now provides strategic consulting and advisory services through GFH Group Inc.

She holds an M.Phil., an MBA and the ICD.D designation. A strong advocate for Canada's digital future, Tracey remains committed to supporting the country's journey toward becoming a fully digital nation.

## Acronyms and Abbreviations

| | |
|---|---|
| ACSS | Automated Clearing Settlement System |
| AML | anti-money laundering |
| ATF | anti-terrorist financing |
| BoC | Bank of Canada |
| CBDC | central bank digital currency |
| CDIC | Canada Deposit Insurance Corporation |
| DeFi | decentralized finance |
| DLT | distributed ledger technology |
| ECB | European Central Bank |
| FCAC | Financial Consumer Agency of Canada |
| FINTRAC | Financial Transactions and Reports Analysis Centre of Canada |
| FRFIs | federally regulated financial institutions |
| L1 | layer 1 |
| MiCA | Markets in Crypto-Assets Regulation |
| OSFI | Office of the Superintendent of Financial Institutions |
| P2P | person-to-person |
| PSPs | payment service providers |
| rCBDC | retail CBDC |
| RPAA | Retail Payment Activities Act |
| RSN | regulated settlement network |
| RTGS | real-time gross settlement |
| RTR | Real-Time Rail |
| SWIFT | Society for Worldwide Interbank Financial Communication |
| wCBDC | wholesale CBDC |

## Executive Summary

Canada's payment system is widely regarded as one of the safest, most resilient and most trusted in the world. It is supported by strong institutions, coordinated infrastructure, prudential regulation and a highly interoperable banking system that collectively reinforce financial stability and public confidence. Over the past two decades, payment modernization in Canada has largely occurred through coordinated evolution involving regulators, major financial institutions and shared infrastructure operators rather than through disruptive market restructuring.

This paper is primarily intended as a predictive analysis of how payments based on distributed ledger technology (DLT) are likely to evolve in Canada given the country's existing market structure, institutional incentives, regulatory dynamics and historical implementation patterns. Alternative approaches are possible, and in some cases may produce greater competition or innovation, but Canada has historically prioritized coordination, interoperability, systemic stability and continuity over disruptive transformation.

## Introduction

DLT introduces capabilities that have the potential to reshape payment infrastructure at a structural level, including tokenization, programmable transactions, shared synchronization of transaction data and atomic settlement.[1] These capabilities create opportunities to modernize treasury management, securities settlement, and domestic and cross-border payments. At the same time, DLT introduces important questions regarding governance, interoperability, market structure, competition, financial stability and monetary sovereignty.

Globally, jurisdictions are pursuing markedly different approaches. The United States has adopted a market-led model centred on private stablecoin issuance and commercial innovation, while Europe is pursuing a regulator-led model anchored in central bank infrastructure and coordinated public policy objectives. Both models reflect broader strategic priorities regarding sovereignty, competition, financial stability and control over digital payment infrastructure.

Canada's path may contain elements of these approaches but will be unique. Canada does not possess the scale advantages of the United States, nor does its institutional structure naturally support a highly centralized European-style framework. Canada's concentrated banking sector, extensive use of shared infrastructure and regulatory emphasis on systemic stability make it more likely that DLT adoption will occur through coordinated evolution.

The concentrated market structure creates both advantages and constraints. Coordination among major institutions is easier than in fragmented markets, supporting interoperability and reducing systemic risk. At the same time, concentrated markets can reduce incentives for disruptive innovation and slow modernization. The implementation history of major payment initiatives in Canada, including the evolution of the Real-Time Rail (RTR), demonstrates the extent to which large-scale infrastructure change requires alignment among incumbent financial institutions, regulators and infrastructure operators.

As a practical matter, the successful implementation of DLT-based payment infrastructure in Canada will require that large financial institutions derive clear strategic or economic value from participation. This reality will shape both the pace of adoption and the architectures most likely to emerge. Models that complement existing institutional structures, regulatory frameworks and settlement infrastructure are therefore more likely to succeed than models that depend upon significant market restructuring or disintermediation of incumbent participants.

The most pragmatic near-term starting point for Canada is the introduction of tokenized deposits issued by regulated financial institutions within the existing prudential and settlement framework. While tokenized deposits will not deliver DLT settlement capabilities, they leverage existing regulation, market trust, banking infrastructure and institutional incentives and offer the lowest-friction path to broad issuer adoption.

1 Please see Appendix A for a glossary of key terms used throughout the paper, to support clarity and consistency, particularly where terminology may be interpreted differently across jurisdictions.

Over time, tokenized deposits, deposit tokens, stablecoins, and potentially wholesale central bank digital currency (wCBDC), could evolve into a federated ecosystem of interoperable networks supported by shared standards and interoperable settlement capabilities. In this environment, multiple forms of digital money could coexist while remaining connected to Canada's broader financial system.

The central challenge for Canada is not the adoption of DLT but rather the governance and coordination required to support meaningful adoption. The long-term outcome for Canada will be shaped less by whether DLT is technically feasible, and more by how standards are developed, how interoperability is achieved, how incentives are aligned and how public policy objectives are balanced against market realities.

Canada's opportunity is not to replicate the United States or Europe, but to leverage its ecosystem to build a coordinated, interoperable and globally connected digital payment framework that preserves trust and stability while enabling evolutionary innovation.

## The Current Payments Model in Canada

Canada's payments ecosystem is the result of decades of intentional design and incremental modernization. It reflects a coordinated architecture in which different components of the system perform distinct but complementary functions, ensuring both efficiency and stability. At its core, the system separates settlement from clearing and user-facing services, while maintaining strong integration across these layers.

At the centre of the system is Payments Canada, which operates the country's core clearing and settlement infrastructure. Lynx, Canada's high-value payment system, provides real-time gross settlement in central bank money and is essential to maintaining financial system stability. Lynx uses the secure Society for Worldwide Interbank Financial Communication (SWIFT) network for transferring domestic and international high-value payments (wire payments) between financial institutions. By ensuring that large-value transactions are settled with finality and without credit risk, Lynx underpins confidence in the financial system and supports the functioning of financial markets. Alongside Lynx, the Automated Clearing Settlement System (ACSS) processes batch-based retail payments, including direct deposits, pre-authorized debits, bill payments and cheques. While highly reliable, ACSS reflects an earlier generation of payment technology, and batch payments lack the speed and data richness increasingly expected by users.

The introduction of the RTR represents a significant step forward for payments in Canada. Designed to enable instant, data-rich payments on a 24/7 basis, the RTR design reflects a uniquely Canadian approach. Rather than utilizing a single exchange model, the RTR has been designed to accommodate multiple exchanges, including third-party networks. While this approach preserves investment in existing exchange capabilities, it may also introduce operational complexity, result in user experiences that vary by exchange, and limit adoption of the RTR exchange. These are aspects to consider as Canada explores a path for DLT.

Overlaying national payment settlement infrastructure is Interac Corporation, owned by a consortium of Canadian financial institutions. Interac is recognized as one of the most reputable brands in Canada (Interac 2025) and is deeply embedded in the daily lives of Canadians through the provision of the Interac Debit and Interac e-Transfer services through its large network of participants. Interac's success is rooted in its ubiquity, bank-grade security, and consistent and reliable user experiences.

The Canadian payment system is supported by a comprehensive regulatory framework. The Department of Finance sets policy direction, the Bank of Canada (BoC) oversees important payment systems, the Office of the Superintendent of Financial Institutions (OSFI) supervises financial institutions, the Financial Transactions and Reports Analysis Centre of Canada (FINTRAC) enforces compliance with anti-money laundering (AML) and anti-terrorist financing (ATF) requirements, and the Financial Consumer Agency of Canada (FCAC) protects consumers (see more on these and other regulators' roles in Appendix B). Together, these entities create a robust environment that seeks to balance innovation and risk management.

Canada's payments ecosystem has historically evolved through a model that is bank-led, regulator-supervised and delivered through shared infrastructure. This approach has produced high levels of trust, interoperability and

systemic resilience, but it has also concentrated influence among incumbent institutions and reduced incentives for disruptive innovation. Major modernization initiatives, including the RTR, open banking, and now stablecoin regulation, demonstrate the extent to which large-scale change in Canada depends upon alignment among large financial institutions, regulators and infrastructure operators.

Alternative policy approaches are possible. More open and competitive market structures can also support trust and safety when supported by appropriate governance and regulatory oversight. However, Canada's historical implementation patterns suggest that policy makers have generally prioritized stability, coordination and institutional continuity over aggressive market restructuring.

# What DLT Changes in Payments

DLT introduces capabilities that are difficult to replicate simultaneously within traditional payment infrastructure, particularly when programmability, shared synchronization of data, interoperability across participants and atomic settlement are considered together. DLT can support the representation of value in digital form through tokenization, enable programmable transactions and reduce reliance on centralized intermediaries in some settlement models. These capabilities extend beyond incremental improvements and introduce the potential for structural change.

DLT enables more data-rich and transparent transactions. Immutable transaction records provide enhanced traceability and auditability, which can improve compliance, enhance fraud detection and prevention, and support more effective oversight. These characteristics make DLT particularly attractive for complex financial transactions involving multiple parties and jurisdictions.

## Programmable Money and Atomic Settlement

One of the most significant features of DLT is the concept of programmable money. Payments can be conditional, automated and integrated into broader business workflows, reducing or eliminating the need for manual intervention, providing traceability and reducing risk. DLT enables new payment use cases such as escrow arrangements, supply-chain finance and automated settlement of contractual obligations. Another feature DLT supports is atomic settlement, which allows for the simultaneous, risk-free settlement of funds. DLT offers a potentially more efficient alternative to aspects of the correspondent banking model currently underpinning cross-border payments. Reducing reliance on intermediary clearing and reconciliation processes can lower settlement time, operational complexity and counterparty risk.

There are several forms of programmable money supported by DLT. Although their features and functionality are similar, each of these digital payment assets is issued, collateralized and regulated differently, with unique implications for the structure, interoperability and functioning of the payment system.

Central bank digital currency (CBDC) represents a direct claim on central bank money in digital form. CBDC can be designed to support retail payment use cases (rCBDC) for individuals and wholesale payment use cases (wCBDC) between payment system participants (banks). The BoC actively investigated DLT for settlement through Project Jasper between 2016 and 2019. Research into an rCBDC began in 2020, but the project was shelved in September 2024 in favour of other payment priorities. In March 2026, the BoC issued a staff paper about Project Samara (Arora et al. 2026), an initiative with the Royal Bank of Canada, the Toronto-Dominion Bank Group and Export Development Canada to evaluate how DLT can improve bond issuance, settlement and life-cycle management. This experiment included atomic settlement using W-CAD, a wholesale Canadian dollar token minted by the BoC.

Tokenized deposits and deposit tokens are both DLT-based representations of commercial bank money and can be issued by regulated banks within the existing prudential framework. Tokenized deposits replicate existing assets on a ledger, while deposit tokens support a new atomic settlement capability.

Tokenized deposits are a digital representation of a traditional bank deposit, recorded on a distributed ledger. When implemented on

proprietary ledgers, they can be transferred between institutions through interoperability mechanisms; however, interbank settlement will generally continue to rely on traditional settlement systems (for example, Lynx or similar infrastructure) and therefore is not inherently atomic. Tokenized deposits are well-suited for retail payment use cases with high-volume flows.

Deposit tokens, by contrast, are native digital instruments typically issued on a shared ledger and designed to function as a transferable settlement asset across participating institutions. While they remain liabilities of the issuing bank, they must be governed by a common rule set. Transactions can settle atomically without reliance on external settlement systems, provided that appropriate governance, liquidity and risk management arrangements are in place. While settlement in central bank money may not be required for such a system to function, settlement on Lynx or using wCBDC would enhance resilience by providing a risk-free settlement asset and a liquidity backstop.

Stablecoins are privately issued digital assets backed by reserves that can be used for payments and other financial applications. Fiat-backed stablecoins are cryptocurrencies backed one-to-one with short-term treasuries and other cash equivalents. As a result of this collateralization approach, stablecoins can be issued by banks and non-bank entities. The use cases for stablecoins have expanded since the coins were introduced over a decade ago to support cryptocurrency trading, and now include wholesale and retail use cases such as corporate treasury management, settlement, cross-border payments, retail payments and inflation hedging. Fintechs and payment service providers (PSPs) can leverage stablecoins as an enhanced infrastructure layer or as an alternative to existing payment options.

### Permissionless Versus Permissioned DLT Solutions

Permissionless, or public ledgers, such as bitcoin and Ethereum, are decentralized, open to anyone and transparent, and can provide high levels of security. Trust is achieved through decentralized consensus mechanisms rather than institutional oversight. Permissioned networks, such as Hyperledger Besu and R3 Corda, restrict access to known, authorized participants who can view, read and/or write to the ledger. These networks can offer greater privacy, scalability and transaction throughput. Their governance is defined by a consortium or central authority, and transaction validation is performed by approved participants. The permissioned option enables stronger identity assurance, privacy and performance, but at the cost of reduced openness and network effects. A hybrid approach is emerging, where permissioned systems handle issuance and compliance, and permissionless networks provide optional interoperability and reach. Permissioned networks are commonly selected in financial service DLT solutions.

## How Are DLT Payments Being Implemented in Other Jurisdictions?

Two reference markets with very different approaches are of interest: Europe and the United States. Europe's policy objectives are to strengthen sovereignty and modernize settlement. The United States' policy objectives are to enable private money innovation and increase currency dominance. These objectives are driving very different implementation approaches, both relevant for Canada.

### Europe's Approach

Europe's approach is regulator driven. The European Central Bank (ECB) is taking steps to ensure that central bank money remains at the heart of the system; tokenized deposits and stablecoins are permitted if they are denominated in euros, meet applicable EU currency and design requirements, and are governed by the European Union.

Europe has been vocal about payments sovereignty. Europe sees CBDC as a way to preserve public money in the digital age and to strengthen European payments autonomy, framing the digital euro as a pan-European retail payment option for online, point-of-sale and offline use. The digital euro project continues to move forward with a target issuance date in 2029 (ECB 2025b, 2025c).

In addition to a focus on its digital euro, the ECB is also creating wholesale DLT settlement capabilities to ensure that tokenized financial markets can

continue to settle in central bank money. Europe is aligning CBDC, tokenized deposits and stablecoins under one policy umbrella and ensuring that all payment tokens have clear rules, but each digital asset operates under different legal classifications: the Markets in Crypto-Assets Regulation (MiCA) for private stablecoins, and traditional banking law for tokenized deposits and CBDCs.

The ECB is introducing a European shared ledger or a European network of interoperable platforms within a broader federated ecosystem governed by shared rules that will link DLT platforms and traditional TARGET Services[2] to enable secure, central bank money settlement for digital asset transactions across the euro area. Projects Pontes and Appia[3] are building the capability for DLT-based asset exchange transactions to settle in central bank money (ECB 2025a, 2025c).

## The United States' Approach

The United States is taking a pro-stablecoin path, as evidenced by House passage of the Anti-CBDC Surveillance State Act and enactment of the GENIUS Act in July 2025.[4] US federal policy has increasingly favoured modernizing payments around private stablecoin issuance and the tokenization of existing commercial bank deposits; public policy is focused on how private digital dollars can scale safely.

Private stablecoins represent an opportunity to support the objective of US dollar dominance. Of the approximately US$313 billion of stablecoins in circulation as of June 30, 99 percent are fiat-backed, and 95 percent of those are US dollar–backed (CoinGecko, n.d.-a). The issuers of US dollar–backed stablecoins have already become meaningful purchasers of short-term US debt, and this is expected to continue. Dollar-backed stablecoins extend US monetary influence in emerging and developing countries. This level of dominance, although it creates risk for US treasury markets, creates an environment in which it has become challenging for stablecoins backed by other currencies and for CBDCs to make significant inroads.

The United States has adopted a market-led model that emphasizes private-sector innovation. Policy makers have discouraged the development of an rCBDC while supporting the growth of stablecoins and other private digital assets. This approach has enabled rapid innovation and reinforced the global dominance of the US dollar, but it has also resulted in a fragmented ecosystem with multiple issuers and platforms.

## External Pressures and Strategic Timing

Canada's transition toward DLT-based payment infrastructure will not occur in isolation. External developments may accelerate domestic adoption timelines and reduce Canada's ability to evolve entirely at its own pace. The rapid growth of US dollar–backed stablecoins, the tokenization of global financial markets and the emergence of cross-border programmable settlement networks could create increasing pressure for Canadian institutions to support interoperable digital payment capabilities.

Large technology platforms, global financial institutions and foreign payment networks are also investing heavily in tokenized financial infrastructure. If these systems achieve meaningful scale internationally before Canada establishes domestic standards and infrastructure, Canadian institutions may increasingly be required to integrate with externally governed networks and standards rather than shaping them domestically.

These pressures introduce strategic considerations beyond payment modernization alone. Questions of monetary sovereignty, domestic control over payment infrastructure, competitive positioning of Canadian financial institutions, and long-term interoperability with global markets may ultimately influence the pace and structure of Canadian DLT adoption as much as domestic policy objectives. Canada's strategic challenge will be to preserve domestic monetary and payment system sovereignty while ensuring interoperability within an increasingly tokenized global financial system.

2 The acronym stands for Trans-European Automated Real-time Gross Settlement Express Transfer and describes financial market infrastructure services "developed and operated by the Eurosystem which ensure the free flow of cash, securities and collateral across Europe" (see www.ecb.europa.eu/paym/target/html/index.en.html).

3 See www.ecb.europa.eu/paym/target/pontes/html/index.en.html; www.ecb.europa.eu/paym/dlt/appia/html/index.en.html.

4 US, Bill HR 1919, *Anti-CBDC Surveillance State Act*, 119th Cong, 2025; *Guiding and Establishing National Innovation for U.S. Stablecoins Act*, 12 USC § 5901 (2025) [*GENIUS Act*].

# What Are the Considerations for Canada?

The experiences of the United States and Europe provide valuable insights into the range of possible approaches to digital money and payments, but neither model is directly transferable. Canada has neither the scale, nor presumably the desire, to replicate or compete with the global dominance of the US dollar, nor does the regulatory environment support full centralization as it does in Europe. DLT can add value for Canada, but the drivers and path to adoption will reflect the structure and requirements of the participants in the Canadian ecosystem.

## The Market

Canada has traditionally looked to the market to drive payment innovation. Canada's financial services market is characterized by high concentration, with the largest six banks holding more than 90 percent of the market share (Toronto Metropolitan University 2024). The "Big Six" exert significant influence over payment systems, necessitating collaboration for innovation. Over the last decade, smaller federally regulated banks, such as EQ Bank and Peoples Bank, and new entrants (fintechs, PSPs, other entities) have made inroads; however, the market remains conservative, prioritizing stability over disruption. Fintechs and other new entrants have advanced in digital product offerings and customer experience enhancements, but they are unlikely, on their own, to drive the national implementation of DLT payments in Canada.

## The Regulators

There are many regulators involved in the oversight of Canada's payments ecosystem, with the BoC emerging as the primary regulator. The BoC's oversight responsibilities have expanded significantly beyond payment systems to include non-bank payment entities. The Retail Payment Activities Act (RPAA) was introduced to bring non-traditional players into a structured, supervised environment.[5] The RPAA was officially launched in September 2025 and continues to be refined. Budget 2025 announced that responsibility for implementation and oversight of consumer-driven banking would transfer from FCAC to the BoC effective April 1, 2026, and Canada's proposed Stablecoin Act was introduced, specifying that the BoC is to manage non-financial institution issuer risk (and likely also wallet provider risk) through RPAA registration and compliance (Government of Canada 2026).

## Regulatory Impacts

If the Department of Finance directed the BoC to issue CBDC, legislative and regulatory changes would be required, including updates to the Bank of Canada Act, the Currency Act and AML/ATF requirements. Depending on CBDC design, amendments might also be required to strengthen privacy laws and know-your-customer requirements, as well as to the RPAA.

Tokenized deposits and deposit tokens are viewed as digital representations of existing bank money and can be accommodated within existing legislation. They are deemed to be regulated by the Bank Act and remain under existing OSFI supervision and subject to Canada Deposit Insurance Corporation (CDIC) protection for CDIC members. Tokenized securities — digital representations of securities assets — remain under the oversight of the Canadian Securities Administrators and the Canadian Investment Regulatory Organization.

The Stablecoin Act defines requirements that apply to all stablecoins issued by non-financial institutions in Canada, including reserve requirements, asset quality, redemption rights, yield and reporting. It focuses on stablecoin issuance, and requires issuers to disclose information about the technology supporting their stablecoins, but does not prescribe the underlying ledger architecture or technology. The Stablecoin Act is also focused on bringing fintechs and other non-financial entities seeking to issue stablecoins under the supervision of the BoC. Federally regulated financial institutions (FRFIs) — including federally regulated banks and federal credit unions — are excluded, as they are already subject to prudential regulation. Bill C-15 received Royal Assent on March 26, 2026, which enacted the Stablecoin Act.[6] The next steps for Canada are to develop specific regulations on

5 *Retail Payment Activities Act*, SC 2021, c 23, s 177.

6 *Budget 2025 Implementation Act, No. 1*, SC 2026, c 3, s 600.

reserve requirements and operational standards, with implementation targeted for 2027.

There have been a few Canadian stablecoin issuer announcements. However, the initial value of the Stablecoin Act will be to require US and foreign issuers of stablecoins to align with Canadian standards.

## Is There a Logical Starting Point?

Table 1 considers each DLT payment option by use case. This high-level analysis demonstrates that no single payment type dominates across use cases. wCBDC and deposit tokens together are particularly well-suited to deliver atomic settlement capabilities. wCBDC and deposit tokens are best suited for bank-to-bank payments, with tokenized deposits also well-suited for some wholesale/infrastructure use cases. Retail use cases can be served by rCBDC, deposit tokens, tokenized deposits and stablecoin. Stablecoin is best suited to support DeFi and crypto trading. There are significant overlaps and key differences between DLT payment assets.

Deposit-taking FRFIs are well-positioned to take advantage of DLT and can issue tokenized deposits and deposit tokens for proprietary use already. If FRFIs wish to exchange these digital assets at scale, this will likely require shared settlement infrastructure operating within a federated ecosystem. Additionally, broad-scale atomic settlement in central bank money would require a central bank digital asset (CBDC or central bank–issued deposit tokens). If a regulated settlement network (RSN) was built in Canada,

Table 1: DLT Payment Option Use Cases

| Use Case | DLT Payment Option | | | | |
|---|---|---|---|---|---|
| | rCBDC | wCBDC | Deposit Tokens | Tokenized Deposits | Stablecoin |
| Central bank money settlement | ✔ | ✔ | ✘ | ✘ | ✘ |
| Atomic settlement | ✔ | ✔ | △* | △ | ✔ |
| Cross-border payments | △ | ✔ | △ | △ | ✔ |
| Interbank fund payments | ✘ | ✔ | △ | △ | △ |
| Monetary policy operations | ✘ | ✔ | ✘ | ✘ | ✘ |
| Treasury management; liquidity optimization | ✘ | ✔ | ✔ | ✔ | △ |
| P2P transfers | ✔ | ✘ | △ | △ | ✔ |
| P2M payments | ✔ | ✘ | △ | △ | ✔ |
| Government disbursements | ✔ | ✘ | △ | △ | △ |
| Financial inclusion | ✔ | ✘ | ✘ | ✘ | △ |
| DeFi activity | ✘ | ✘ | △ | △ | ✔ |
| Crypto trading | ✘ | ✘ | △ | △ | ✔ |
| Currency hedge | ✘ | ✘ | ✘ | ✘ | ✔ |

Key: ✔ Well-suited △ Possible ✘ Ill-suited

*Source:* Author analysis.
*Notes:* DeFi = decentralized finance; P2M = person-to-merchant; P2P = person-to-person.
*CBDC/tokenized central bank money is required for atomic settlement in central bank money.

either as a public utility or by the banks themselves, FRFIs could issue stablecoins natively to the RSN that would support atomic settlement.

FRFIs are not likely to issue digital assets on public, permissionless ledgers, as assets issued on these networks may not qualify for Group 1 treatment under OSFI's crypto-asset framework. Group 2 assets are subject to significantly more conservative capital treatment. The Stablecoin Act does not currently prescribe whether stablecoins must be issued on public or private ledgers, providing issuers with flexibility in their choice of ledger architecture. The Stablecoin Act defines requirements for collateralization and reporting but is silent on ledger technology.

Although the regulatory environment already contemplates digital asset issuance by FRFIs, their ability to realize the full value of DLT may remain constrained until Canada builds a private, permissioned layer 1 (L1) blockchain and the BoC supports atomic digital settlement. Canada's large banks have experience in building payment systems (Interac Debit and e-Transfer) and could build a proprietary blockchain to support digital asset exchange. Existing real-time gross settlement (RTGS) infrastructure (Lynx, and the RTR when it launches) can provide an interim settlement mechanism until the BoC supports digital settlement.

Private stablecoin issuers have technology flexibility and can deliver the benefits of DLT through public chains. Private issuers must meet RPAA requirements, as well as the stringent collateralization and reporting obligations as defined by the Stablecoin Act. Private stablecoins may not be natively interoperable and may rely on third parties to support broad acceptance. The utility and adoption of privately issued stablecoins will depend on the use case and the network.

## Can Canada Avoid Fragmentation?

DLT solutions must preserve the trust, stability and reliability that characterize Canada's existing payment framework. Although market-driven, Canada values much of what Europe is focused on building. Policy objectives and the target end state should be defined early to ensure Canada achieves the desired outcome.

The OSFI crypto-asset guidelines may discourage FRFIs from issuing digital assets on a public L1 ledger. Individual FRFIs are already investigating the value of digital assets on private permissioned ledgers. Each FRFI has been selecting its own ledger provider based on proprietary requirements. Rather than each FRFI investing in proprietary solutions that do not support interoperability, Canada could evolve toward shared foundational settlement infrastructure within a federated ecosystem that could deliver significant operational efficiencies for proprietary FRFI digital asset use cases.

One possible architecture within a federated ecosystem of interoperable networks would involve the BoC operating a private, permissioned settlement network. Such a ledger architecture could support the issuance of CBDC or central bank–issued deposit tokens, which, although not currently a priority for the BoC, should be considered as a possibility in the future. A partitioned BoC ledger could allow FRFIs to participate directly to support both proprietary and interbank use cases. Interbank transactions would benefit from the availability of CBDC or BoC-issued deposit tokens on the same ledger to provide instant, legally binding atomic settlement. This is the structure used to support Project Samara. Although technically efficient, this structure introduces legal and operational complexity (Arora et al. 2026).

The second option is for the BoC to focus only on providing atomic settlement in tokenized central bank money for digital assets within a federated ecosystem of interoperable networks. In this scenario, the BoC would build and operate a private, permissioned settlement ledger, and FRFIs would develop and operate digital asset solutions on their own private, permissioned networks. Interoperability mechanisms could provide connectivity between FRFI networks, and the BoC ledger would provide a common settlement layer in tokenized central bank money. This structure is similar to the system architecture being designed and implemented in Europe and may represent a more likely path for Canada.

Even without digital central bank money settlement, there is an opportunity for FRFIs to build a private, permissioned and partitioned

ledger that leverages RTGS settlement on Lynx or the RTR. FRFIs could use this shared ledger to support more efficient issuance of proprietary digital assets and also as a shared platform supporting an interoperable multi-bank deposit token, or even a jointly issued stablecoin. This solution would require tightly defined technology specifications for the ledger and development of token standards to support interoperability. Operational oversight and system governance would need to be carefully managed. A FRFI joint ledger solution could be operated by bank-owned Interac Corporation, an organization with experience in developing common technical standards and rules and governing shared systems, as demonstrated by Interac Debit and e-Transfer.

Given the significant market share of Interac owner banks, it is prudent to require involvement from public authorities to ensure that significant shared digital asset solutions support systemwide stability, neutrality and alignment with broader policy objectives. As is the case with Interac Debit and e-Transfer, if FRFIs introduce an Interac-branded shared digital asset solution, it should be considered for designation as a prominent payment system and overseen by the BoC.

The central challenge for Canada is unlikely to be technical implementation. The more difficult challenge will be governance. Interoperability across networks, standards development, settlement design, competitive neutrality, operational oversight and control of shared infrastructure all introduce questions regarding institutional authority and incentive alignment. These issues become increasingly complex when incumbent institutions, new entrants, regulators and public policy objectives do not fully align.

Canada has historically managed this complexity through coordinated governance involving regulators and large financial institutions. Whether this approach remains sufficient in a DLT environment — where infrastructure, settlement and digital asset issuance become more tightly integrated — remains uncertain. Coordination may accelerate adoption and interoperability, but it may also slow innovation and reinforce existing market concentration.

A federated ecosystem of interoperable networks would also require governance frameworks that extend beyond technical interoperability alone. Questions surrounding participant access, liability allocation, operational resilience, dispute resolution, settlement finality, standards development and competitive neutrality would need to be clearly defined. Existing Canadian models suggest governance would likely involve some combination of regulators, infrastructure operators and major financial institutions, with the BoC focused on systemic risk and settlement oversight and industry participants responsible for operational delivery and standards implementation. Whether such a framework can balance interoperability, innovation and competitive fairness without reinforcing existing market concentration remains an open question.

## Privately Issued Stablecoins

There are currently four privately issued Canadian stablecoins (CADC, CADD, CADX and QCAD). Currently, although it can be highly variable, the daily trading volume for the three publicly traded coins (CADX excluded) typically totals less than CDN$100,000.[7] As more private companies issue stablecoins, trading volume is likely to increase.

Canadian stablecoins can operate across multiple independent blockchains, which provides utility for users. However, these tokens do not natively interoperate with one another, which introduces operational complexity and may impact adoption. End-user management of multiple stablecoins is simplified through the availability of wallets linked to crypto bridges that can move assets between chains and convert value from one stablecoin to another.

There are meaningful niche and specialized use cases for privately issued stablecoins, particularly in cross-border payments, treasury management, DeFi and crypto-asset trading. However, the structure of the Canadian market may make it difficult for fragmented private issuers to achieve broad domestic scale, especially if coordinated bank-led solutions emerge.

The competitive dynamic for tokenized payments in Canada will be shaped by how the FRFIs move forward with interbank solutions and by the availability of digital settlement in central bank money. New entrants that opt to issue

7 For a specific example, using data from the independent crypto data aggregator CoinGecko (n.d.-b) for July 22, 2026, the combined 24-hour trading volume for CADC, CADD and QCAD was US$45,702.94.

fiat-backed stablecoins are likely to need a very compelling or niche user experience for success.

## Conclusion: What Is Canada's Desired Outcome?

Canada's transition toward DLT-based payment infrastructure is unlikely to occur through abrupt disruption or the rapid displacement of existing institutions. Historical implementation patterns, regulatory structures and market concentration suggest that adoption will occur incrementally through coordinated evolution involving regulators, large financial institutions and shared infrastructure operators.

This approach creates both strengths and constraints. Canada's concentrated market structure supports interoperability, systemic stability and coordinated implementation, but it may also slow innovation, reduce competitive pressure and limit the pace of transformative change. These tensions are likely to shape major decisions regarding digital money architecture, governance, settlement design and market access.

No single DLT payment model currently satisfies all policy, operational and commercial objectives. Tokenized deposits, deposit tokens, stablecoins and CBDC each address different use cases and introduce different trade-offs relating to programmability, settlement finality, governance, interoperability and monetary control. As a result, Canada is more likely to evolve toward a federated ecosystem than toward a single unified platform.

In the near term, a likely path is the gradual introduction of tokenized commercial bank money operating alongside existing settlement infrastructure such as Lynx and the RTR. Over time, if market demand, regulatory alignment and institutional incentives converge, these systems may evolve toward more advanced atomic settlement models supported by shared standards and potentially wholesale central bank digital money.

Canada's opportunity is not to replicate either the United States or Europe. It is to determine how digital money can evolve within a financial system that prioritizes trust, resilience, coordination and institutional stability while remaining sufficiently open to support meaningful innovation and global competitiveness.

The long-term outcome may depend less on the technical capabilities of DLT and more on governance. Absent meaningful coordination, Canada risks evolving toward fragmented proprietary networks with increasing dependence on externally governed payment infrastructure. Questions surrounding interoperability, standards, competitive neutrality, regulatory oversight, settlement design and control of shared infrastructure will determine whether Canada develops a coherent digital payments ecosystem or a fragmented collection of proprietary networks.

### AI Disclosure

*Artificial Intelligence Tool:* Chat GPT; *Writing — Review & Editing:* ChatGPT (GPT-5.6 Sol) was used to assist with a factual review of the final draft of this paper, including identifying statements requiring verification, clarification or qualification. All substantive analysis, interpretations, conclusions and recommendations are those of the author. The author independently reviewed the issues identified through this process, verified relevant source material where appropriate, and retained responsibility for all final editorial decisions and the accuracy of the published work.

# Appendix A: Glossary of Key Terms

**Atomic settlement:** The simultaneous and irrevocable exchange of assets and funds within a single transaction, eliminating settlement risk and the need for reconciliation between parties.

**Automated Clearing Settlement System (ACSS):** Canada's legacy batch-based retail payment system operated by Payments Canada, used for processing payments such as direct deposits, pre-authorized debits and cheques.

**Central bank digital currency (CBDC):** A digital form of central bank money that represents a direct claim on the central bank. CBDC may be designed for retail use (rCBDC) by individuals and businesses or wholesale use (wCBDC) by financial institutions and settlement between payment system participants.

**Deposit tokens:** Digital representations of commercial bank money issued on a shared, permissioned ledger, governed by a common rule set. Deposit tokens are designed to function as transferable settlement assets between institutions and can support atomic settlement.

**Distributed ledger technology (DLT):** A technology that enables the recording, sharing and synchronization of data across multiple nodes in a network. DLT supports tokenization, programmability, and decentralized or distributed validation of transactions.

**Federated ecosystem of interoperable networks:** An architectural and governance model in which multiple distributed ledger networks, settlement systems and digital asset platforms operate independently while remaining connected through shared standards, interoperability mechanisms and coordinated governance frameworks.

**Fiat-backed stablecoin:** A privately issued digital asset designed to maintain a stable value by being backed one-to-one with fiat currency or high-quality liquid assets such as short-term government securities.

**Interac Corporation:** Interac is a Canadian payment network owned by a consortium of financial institutions, providing widely used services such as debit payments and P2P transfers (for example, Interac e-Transfer).

**Layer 1:** A layer 1 (L1) blockchain is the foundational base network architecture (for example, Ethereum) that operates independently, handles its own consensus and security, and processes and finalizes transactions without relying on another network.

**Ledger of ledgers:** A specific interoperability architecture within a broader federated ecosystem of interoperable networks, where multiple independent ledgers are connected through shared standards or interoperability layers.

**Lynx:** Canada's high-value payment system operated by Payments Canada. Lynx enables real-time gross settlement (RTGS) in central bank money and is critical to financial system stability.

**Permissioned DLT:** A type of distributed ledger where access is restricted to approved participants. Governance, validation and data visibility are controlled by a defined group of entities, typically used in financial services.

**Permissionless DLT:** A public distributed ledger that is open to participation by anyone, including to validate transactions and to access data. Examples of permissionless DLTs include blockchain networks such as bitcoin and Ethereum.

**Programmable money:** Digital money that can be embedded with rules or conditions, enabling automated execution of transactions based on predefined logic (for example, smart contracts).

**Real-Time Rail (RTR):** Canada's forthcoming real-time payment system designed to enable instant, data-rich payments on a 24/7 basis, supporting enhanced speed, transparency and functionality.

**Retail central bank digital currency (rCBDC):** A form of CBDC designed for use by the general public that acts as the digital equivalent to physical cash.

**Retail Payment Activities Act (RPAA):** Canadian legislation that establishes a supervisory framework for PSPs, administered by the BoC.

**Settlement (payment context):** The final transfer of funds between financial institutions that completes a payment transaction and eliminates counterparty risk.

**Shared ledger:** A single distributed ledger infrastructure on which multiple participants issue, transfer and settle digital assets under a common governance and rule framework.

**Stablecoin Act (Canada):** Canadian legislation establishing requirements for stablecoin issuance, including reserve backing, redemption rights and regulatory oversight, particularly for non-bank issuers.

**SWIFT:** The Society for Worldwide Interbank Financial Telecommunication (in full) is a global messaging network that enables more than 11,000 institutions/businesses around the globe to securely exchange payment instructions and other financial information.

**Tokenization:** The process of representing real-world assets or financial claims (such as deposits or securities) in digital form on a distributed ledger.

**Tokenized deposits:** Digital representations of traditional bank deposits issued on a distributed ledger, typically on proprietary or institution-specific platforms. Interbank settlement generally continues to rely on existing payment systems.

**Wholesale central bank digital currency (wCBDC):** A form of CBDC designed for use by financial institutions to settle high-value transactions, often envisioned as a supporting asset for DLT-based financial market infrastructure.

*Source*: Compiled by author.

# Appendix B: Financial Services Regulators in Canada

## Federal Financial Regulators

| Regulator | Core Responsibility and Mandate |
|---|---|
| **Office of the Superintendent of Financial Institutions (OSFI)** (www.osfi-bsif.gc.ca/en) | **Prudential oversight:** solvency and safety of federal banks, trust/loan companies, insurance firms and private pension plans |
| **Financial Consumer Agency of Canada (FCAC)** (www.canada.ca/en/financial-consumer-agency.html) | **Consumer protection:** enforces compliance with consumer laws, regulates retail banking practices and promotes national financial literacy |
| **Bank of Canada (BoC)** (www.bankofcanada.ca/) | **Monetary system and stability:** sets interest rates, provides system liquidity, acts as "lender of last resort" and oversees systemic clearing and payment infrastructures |
| **Canada Deposit Insurance Corporation (CDIC)** (www.cdic.ca/) | **Deposit insurance and bank resolution:** insures eligible deposits up to $100,000 per category if a bank fails and manages the restructuring or orderly closure of failed member institutions |
| **Financial Transactions and Reports Analysis Centre of Canada (FINTRAC)** (https://fintrac-canafe.canada.ca/) | **Anti-money laundering (AML):** detects, prevents and deters money laundering and terrorist financing across all financial institutions, money service businesses and real estate sectors |

## Provincial and Territorial Regulators

| Regulator | Core Responsibility and Mandate |
|---|---|
| **Provincial securities commissions** (for example, the Ontario Securities Commission, the Autorité des marchés financiers, the BC Securities Commission) | **Investment and market regulation:** regulates capital markets, stock exchanges, public company disclosures, and registers investment firms and advisers within their respective provinces |
| **Canadian Securities Administrators (CSA)** (www.securities-administrators.ca/) | **National coordination umbrella:** non-governmental body that harmonizes and cross-coordinates securities regulations across all 13 provinces and territories |
| **Provincial credit union and insurance regulators** (for example, the Financial Services Regulatory Authority of Ontario) | **Provincial institutions:** regulates credit unions (*caisses populaires*), provincial insurance firms, mortgage brokerages and loan/trust companies operating locally |

## Industry Self-Regulatory Organizations

| Regulator | Core Responsibility and Mandate |
|---|---|
| **Canadian Investment Regulatory Organization** (www.ciro.ca/) | **Investment dealers and mutual funds:** formed by the amalgamation of the IIROC (Investment Industry Regulatory Organization of Canada) and the MFDA (Mutual Fund Dealers Association of Canada); directly oversees investment dealers, mutual fund dealers and trading activity on Canadian marketplaces |

*Source*: Compiled by author.

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